# Namma Space: Intelligent Spatial Engine & Photorealistic 3D Digital Twin

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## 1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

Physical indoor spaces (megastores, corporate headquarters, academic campuses, hospitals) present severe navigation and spatial orientation bottlenecks. Traditional static 2D blueprints fail to convey spatial volume, vertical clearance, or dynamic obstacle layouts. Existing industrial digital twin pipelines either mandate prohibitively expensive LiDAR surveying rigs (>$10,000) or produce non-navigable radiance fields that fail on consumer devices.

**Namma Space** introduces an automated, end-to-end spatial computing engine that converts ordinary smartphone camera video and inertial sensor streams into **photorealistic, millimeter-accurate 3D digital twins** and **autonomous robotics navigation costmaps** without auxiliary surveying gear. Delivered through a zero-install WebGL/WebXR client, it features desktop first-person FPS roaming (WASD), touch-optimized mobile virtual joysticks, 2D floor radar, and obstacle-aware A* pathfinding.

## 2. DUAL-STREAM END-TO-END SYSTEM ARCHITECTURE

Monocular 3D spatial reconstruction suffers from a fundamental tension between **visual photorealism** and **geometric watertightness**: radiance fields (3DGS) provide view-dependent lighting but lack physical collision meshes, while classical meshing produces watertight bounds but suffers from melted-wax artifacts on flat surfaces. Namma Space establishes a synchronized **Dual-Stream Architecture** tied to a unified metric world coordinate frame:

PIPELINE SUBSYSTEM MATRIX // CRA003 ARCHITECTURE

| SUBSYSTEM | PRIMARY TECHNOLOGY | TARGET HARDWARE | PRIMARY OUTPUT ARTIFACT |
|---|---|---|---|
| **Sensor Ingestion** | PoseCam (ARCore 6-DoF VIO + 200 Hz STMicro IMU) | Android Smartphone | `poses.csv`, `imu.csv`, `frames/*.jpg` |
| **Quality Gates** | Laplacian Variance Blur Filter (Var > 100) | Host CPU (Python / OpenCV) | Decimated Keyframe Trajectory Sequence |
| **Neural Metric Depth** | Depth-Anything-V2 Metric Indoor ViT | Laptop GPU (CUDA / Vulkan) | Scale-consistent metric depth maps (meters) |
| **Geometric TSDF Fusion** | Open3D ScalableTSDFVolume (3cm voxel / 9cm band) | Host CPU / 32 GB RAM | Watertight volumetric isosurface & planes |
| **Champion Representation** | Model 1: Dense Metric Point Cloud (2.81M points) | WebGL 2.0 (Single Draw Call) | Photorealistic 3D Digital Twin (60 FPS) |
| **Robotics Navigation** | Elevation Slicing [0.08m, 1.80m] + A* Obstacle Dilation | ROS 2 Nav2 Map Server | `floormap.png`, `floormap.yaml` |

## 3. FIELD CAPTURE STANDARD OPERATING PROCEDURE (SOP)

Reconstruction fidelity in deep monocular SLAM depends critically on baseline parallax and frame sharpness. We formulate a mathematically repeatable SOP executed on consumer smartphones:

- **Sensor Configuration:** 1080p/4K at 60 FPS. Auto-Exposure and Auto-Focus are hard-locked (AE/AF Lock) to prevent luminance shifts and focal scale breathing across frames.
- **Perimeter-Loop Pattern:** The surveyor executes a smooth heel-to-toe gait along the room perimeter, angling the optical axis 45° inward toward the room centroid to maintain continuous parallax across wall-floor junctions.
- **360° Island Orbits:** Center-room obstacles (islands, conference tables, structural columns) are circumscribed with an inward-facing orbit at 1.2 m standoff distance.
- **Loop Closure Mandate:** The terminal trajectory re-traverses the initial coordinate origin for ≥3.0 seconds, providing pose-graph constraints that neutralize cumulative drift.
- **Automated Blur Rejection:** Blurry frames are rejected via Laplacian variance: $\text{Var}(\nabla^2 I) = (1/WH) \sum (\nabla^2 I(x,y) - \mu_{\nabla^2 I})^2 > 100.0$. Subsequent keyframe decimation operates on camera displacement ($|\Delta t| > 0.15$m or $|\Delta\theta| > 8°$).

## 4. MATHEMATICAL FOUNDATIONS & COORDINATE FRAMES

### 4.1 Pinhole Unprojection into Metric Camera Frame

Given pixel coordinate (u, v) with estimated metric depth z (in meters) and camera intrinsics (fx, fy, cx, cy):

```
X_c = ((u - cx) * z) / fx, Y_c = ((v - cy) * z) / fy, Z_c = z P_c = [X_c, Y_c, Z_c, 1]^T
```

### 4.2 Coordinate Frame Conversion: OpenGL to OpenCV

ARCore coordinates follow OpenGL convention (+X right, +Y up, -Z optical forward). The SLAM pipeline follows OpenCV (+X right, +Y down, +Z forward). We map poses via diagonal transformation matrix:

```
T_GL_TO_CV = diag(1.0, -1.0, -1.0, 1.0) T_world_cv = T_world_gl · T_GL_TO_CV P_world = T_world_cv · P_c
```

### 4.3 Hardware Gravity Vector Lock (Eliminating Drift & Tilt)

Monocular SLAM without gravity constraints designates Frame 1 as the world origin (0,0,0). A 25° wrist pitch bakes a permanent 25° diagonal tilt into the room, while turns drift into banana curves. PoseCam locks the world frame directly to **Earth's hardware gravity vector** $g = [0, -9.81, 0]^T$ via the phone's 200 Hz STMicro accelerometer, guaranteeing the floor is clamped horizontal at Y = 0.00 m.

## 5. MODEL 1 CHAMPION SELECTION & COMPARATIVE ANALYSIS OF 10 FORMATS

To rigorously determine the optimal 3D representation for indoor roaming, we generated 10 distinct models from the same sensor capture. **Model 1 (Dense Metric Cloud)** emerged as the unequivocal champion for the competition submission.

**CHAMPION: Model 1 Dense Metric Point Cloud (2,810,480 Points)**

**Architectural Superiority:** Unprojects Depth-Anything-V2 metric depth directly into fused coordinates without forcing synthetic polygonal stitching across empty spaces.

- **100% Photorealistic RGB:** Captures fine whiteboard text, ceiling fan blades, chair legs, and power cables without triangle blur.
- **0% Triangular Face Warping:** Eliminates erroneous triangular bridges stretched across open doors and aisles.
- **Locked 60.0 FPS WebGL:** Native single-draw-call buffer geometry with low GPU memory consumption.

**Comparative Evaluation of the 9 Rejected Models**

| # | REPRESENTATION MODEL | PIPELINE METHOD | PRIMARY FAILURE MODE / WHY REJECTED |
|---|---|---|---|
| 02 | **Trimmed Poisson Mesh** | Screened Poisson Isosurface | Over-smooths wall corners; creates artificial bubbles on unobserved door backs. |
| 03 | **Ball-Pivoting Mesh (BPA)** | Rolling-Sphere Triangulation | Leaves triangular pinholes and perforated meshes on low-texture painted walls. |
| 04 | **Alpha-Shapes Hull** | Concave Polyhedral Alpha Hull | Radius parameter bridges empty corridors between adjacent student desks. |
| 05 | **TSDF Marching Cubes** | Zero-Crossing Voxel Extraction | 3cm voxel discretization blurs sharp monitor bezels, power outlets, and fan rods. |
| 06 | **Planar CAD BIM Shell** | Multi-Model RANSAC Fitting | Isolates perimeter walls but discards interior furniture, computers, and whiteboard. |
| 07 | **3D Voxel Lattice** | Discrete Octree Grid ($5cm^3$) | Blocky voxel artifacts suitable for drone collision, but visually non-photorealistic. |
| 08 | **LingBot-Map GCT** | Curvature Tensor Cloud | Encodes Gaussian/Mean curvature for segmentation, losing natural RGB realism. |
| 09 | **CAD Bounding Boxes** | Oriented Bounding Box Cluster | Abstract wireframes only; lacks surface visual representation. |
| 10 | **Oriented Surfels Disks** | Tangent Surface Disks | Incurs 3× higher draw call overhead in WebGL compared to continuous point clouds. |

## 6. SPATIAL DISCRETIZATION & 2D ROS NAVIGATION COSTMAPS

Continuous 3D spatial reconstructions are converted into safe, non-colliding navigation paths through explicit geometric occupancy modeling.

### 6.1 Geometric Vertical Height Bandpass Slicing: [0.08m, 1.80m]

To prevent erroneous obstacle registration, points are filtered within a strict vertical elevation band:

- **Ground Noise Exclusion (h < 0.08 m):** Door sills, low carpets, floor trim, and power cords are classified as walkable floor datum (y = 0.00 m).
- **Overhead Fixture Exclusion (h > 1.80 m):** Hanging chandeliers, ceiling fans, structural beams, and HVAC ducting are eliminated.
- **True Collision Obstacle Band (0.08m ≤ h ≤ 1.80m):** Student desks, chairs, walls, structural pillars, and cabinets are mapped as occupied cells.

### 6.2 2D ROS Costmap Generation & Navigation Server Specification

Filtered obstacle points are orthographically projected onto a 2D discrete grid with a metric resolution of s = 0.05 m/pixel (5 cm grid cells). Occupancy values are mapped to standard ROS 2 Nav2 format:

```
# floormap.yaml (ROS 2 Nav2 Map Server Spec) image: floormap.pgm resolution: 0.050000 origin: [-4.250000, -3.100000, 0.000000] negate: 0 occupied_thresh: 0.65 free_thresh: 0.196
```

### 6.3 Obstacle Dilation & Catmull-Rom A* Pathfinding

To account for robot / avatar collision volume, occupied costmap cells are dilated by an avatar clearance radius r_clear = 0.30 m. A* calculates the optimal shortest path across collision-free cells, which is interpolated into a continuous 3D navigation floor ribbon using Catmull-Rom cubic splines.

## 7. ZERO-INSTALL WEB CLIENT & MOBILE WEBXR AR

The platform client runs purely client-side across mobile and desktop browsers with zero installation:

- **Rendering Engine:** Three.js rendering 2.81M points at a locked **60.0 FPS** on standard integrated laptop GPUs (Intel Iris Xe / Apple Silicon) and mobile browsers.
- **Dual Controls:** Desktop FPS PointerLock (WASD, Shift sprint at 3.4 m/s, Space jump, C crouch, procedural footstep audio) and mobile virtual touch joysticks (MOVE + LOOK).
- **Live Floor Radar HUD:** Real-time synchronized 2D floor radar displaying player coordinates, speed, and heading orientation.
- **Client-Side Drag & Drop:** Allows testing local .ply models in-browser without server upload roundtrips.

## 8. EXPERIMENTAL BENCHMARKS & SYSTEM VERIFICATION

| EVALUATION METRIC | TESTED SYSTEM VALUE | MEASUREMENT STANDARD / ENVIRONMENT |
|---|---|---|
| **Web Viewer Frame Rate** | **60.0 FPS (Locked)** | Chrome 128 / Intel Iris Xe & Nvidia RTX 3050 Laptop |
| **Reconstruction Turnaround Time** | **3 min 45 sec** | 185 keyframes → Depth-Anything-V2 + TSDF + 10 Models |
| **Metric Scale Accuracy** | **< 2.1% across 12 m span** | Bosch laser rangefinder vs ARCore metric pose trajectory |
| **Ground Plane Level Flatness** | **Δy < 0.02 m (Flat)** | Hardware gravity locked floor datum vs monocular tilt |
| **A* Pathfinding Compute Latency** | **12 ms** | 250 × 180 Navgrid (r_clear = 0.30 m) |

## 9. COMPETITION DELIVERY ROADMAP (ROUNDS 1, 2, 3)

- **Round 1 (Due 31 Oct):** Field Capture SOP, 60 FPS Web Roaming Twin, 5-Page Technical Abstract PDF, 3–5 min Video Walkthrough.
- **Round 2 (Due 30 Nov):** OWLv2 Open-Vocabulary 3D Semantic POI Tagging, Multi-Room Path Router, Spatial Keyword Search UI.
- **Round 3 (16–17 Dec @ IIT Bombay):** On-Site Venue Capture (Lecture Hall Complex - LHC), Rapid-Turnaround Digital Twin, Live Jury Pitch, and **Mobile WebXR Live Camera Guidance** (Juspay Hiring Track).